

Monday Warm Up: Unit 3, Magnetic Forces and Fields

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March 2, 2026

1 Memory Bank

- $\hat{i} \times \hat{j} = \hat{k}$, $\hat{j} \times \hat{k} = \hat{i}$, $\hat{k} \times \hat{i} = \hat{j}$... The direction of the *cross-product* follows this pattern. Reversing the order of any two vectors introduces a minus sign.
- $\hat{i} \times \hat{i} = 0$, $\hat{j} \times \hat{j} = 0$, $\hat{k} \times \hat{k} = 0$... The *cross-product* is zero when both vectors are parallel.
- $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}|\sin\theta$... The magnitude of the cross-product of two vectors $\vec{a}$ and $\vec{b}$, given the angle θ between them.
- $\vec{F} = I\vec{L} \times \vec{B}$... The Lorentz force, by a magnetic field $\vec{B}$ on a *current* i of length and direction $\vec{L}$.
- $\vec{F} = q\vec{v} \times \vec{B}$... The Lorentz force, by a magnetic field $\vec{B}$ on a *charge* q with velocity $\vec{v}$.

2 The Cross-Product

1. Let $\vec{a} = 4\hat{i} + 4\hat{j}$, and $\vec{b} = -4\hat{i} + 4\hat{j}$. Calculate $\vec{a} \times \vec{b}$. What is the angle between the two vectors?

3 Magnetic Force on Currents and Charge

1. A 0.750 meter section of cable carrying current to a car starter motor makes an angle of 60 degrees with the 5.50×10^{-5} T magnetic field of the Earth. What is the current when the wire experiences a force of 7.00×10^{-3} N?

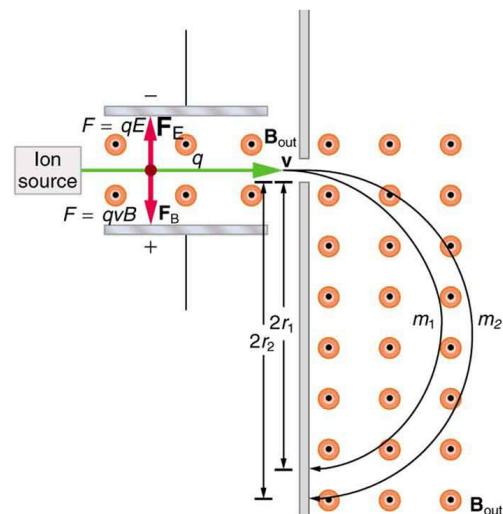


Figure 1: The basic structure of a mass spectrometer.

the positive ions must be $v = E/B$.

3. The positive ions leave the E-field and enter the right side of the structure. The B-field remains uniform, and out of the page. Set the centripetal force equal to the Lorentz force to derive an expression for the radii r_1 and r_2 .

4. Suppose we know that $r_2 = 25$ cm, and corresponds to hydrogen ions. What is r_1 , if those ions are heliums with +1 charge (missing one electron)?

2. Consider Fig. 1. In the upper left portion of the device, a structure with capacitance provides a uniform E-field upwards, and there is a uniform B-field out of the page. Show that if $F_B = F_E$, that the velocity of